3 **(a)** Define momentum.

..... [1]

(b) Fig. 3.1 shows two toy trains, A and B, on a track.

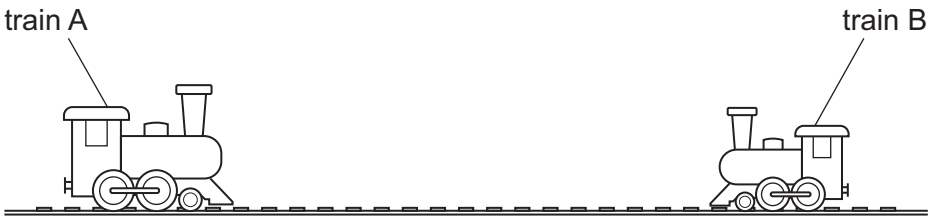

Fig. 3.1

The mass of train A is 0.45 kg and the mass of train B is 0.21 kg. The trains do **not** have motors.

Train A travels towards train B and they collide.

Immediately before the collision:

- the velocity of train A is 0.34 m / s to the right
- the velocity of train B is 0.12 m / s to the left.

The trains stick together when they collide.

Calculate the velocity of the trains immediately after the collision.

velocity = ...

direction .....

[4]

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(c) A different train is travelling with a momentum of 0.26 kg m / s.

The train slows down and stops after 2.1 s.

Calculate the average force acting on the train during this time.

average force = ... [2]

[Total: 7]

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